MPC Heterogeneity and Targeted Transfers with Limited Information: A Quantitative Analysis

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Abstract

Fiscal transfers are more effective when targeted at households with high marginal propensities to consume (MPCs), but MPCs depend heavily on unobservable preference parameters such as discount factors. I study the design of consumption-maximizing transfers in a standard Aiyagari (1994) model with discount factor heterogeneity, comparing four regimes that differ in the government's information set: a first best where all household states are observed, targeting on assets and income, targeting on income alone, and uniform transfers. I define a *targeting ratio* that measures the fraction of first-best gains each regime captures over uniform transfers. In the baseline calibration, income targeting captures only about 23% of these gains. Conditioning on both assets and income raises the ratio to roughly 89%. The concavity of the consumption function also shapes the informational problem: as the transfer budget grows, the targeting ratio for income targeting tends to fall, so larger transfers further widen the gap between income targeting and the first best.

1 Introduction

The effectiveness of fiscal transfers as a tool for aggregate demand stabilization depends on the marginal propensity to consume (MPC) of recipient households. If the government could perfectly identify and target high-MPC households, transfers would generate larger consumption responses per dollar spent. This insight motivates a large body of research documenting MPC heterogeneity and exploring the effect of targeted transfers (Oh and Reis, 2012; Jappelli and Pistaferri, 2014; Aguiar et al., 2024).

A central difficulty is that the household characteristics most responsible for MPC variation may be unobservable to the government. Jappelli and Pistaferri (2014) document

an average MPC of 48% with substantial heterogeneity related to cash on hand using Italian survey data. Jappelli and Pistaferri (2020) use Italian panel data and show that controlling for unobserved household fixed effects halves the estimated consumption gains from targeted transfers, implying that a large share of the policy-relevant MPC variation reflects preferences rather than observable circumstances. Gelman (2021) decomposes MPC variance into a persistent-characteristics component (proxying for discount factor heterogeneity) and a temporary-circumstances component (driven by transitory shocks), finding that the two contribute roughly equally. Fagereng et al. (2021) find that liquid wealth is the strongest observable predictor of the MPC using Norwegian lottery data. Crawley and Kuchler (2023) confirm, using Danish administrative data, that liquid wealth predicts the MPC well across all household characteristics they examine. More recently, Lewis et al. (2026) estimate that standard observables such as income, age, and education account for roughly 8% of MPC heterogeneity using US stimulus payment data, though liquid wealth is not available in their main analysis.[1] Taken together, these findings point to a gap between the MPC variation that exists in the population and the share that a government can exploit using the information typically available to it. This paper asks how large that gap is.

To answer this question, I use a standard incomplete-markets model à la Aiyagari (1994) with ex-ante heterogeneity in discount factors. Discount factor heterogeneity is a parsimonious way to generate the large dispersion in MPCs and wealth holdings observed in the data (Carroll et al., 2017; Kaplan and Violante, 2022), and it creates a direct link between unobservable preferences and the MPC that makes the government's information problem nontrivial. In this environment, two households with identical observable income y can have very different MPCs because they differ in their discount factor β and in their endogenous asset position a, which the government may or may not observe. The government must allocate a fixed lump-sum transfer budget M across the population to maximize the aggregate contemporaneous consumption response.[2]

I compare four regimes in partial equilibrium, ordered by the government's information set. In the *first best*, the government observes the full household state (a, y, β) and chooses transfers $t(a, y, \beta)$. Under *asset-and-income targeting*, the government observes (a, y) but not the discount factor, and chooses $t(a, y)$. Under *income targeting*, the government observes

[1] Earlier work in development economics regarding the practical challenges of targeting transfers is Paul (1992), who notes that governments in developing countries often cannot observe household income.

[2] The optimal taxation literature following Mirrlees (1971) and Saez (2001), and more recent quantitative work such as Heathcote et al. (2017), studies the design of tax and transfer schedules that balance redistribution against efficiency costs. The present paper instead quantifies the design of stimulus transfers: it takes the planner's objective and the stationary distribution as given, and measures how much of the first-best consumption response survives when the planner conditions on coarser information sets.

only y and chooses $t(y)$. Finally, under *uniform transfers*, every household receives $t = M$. To compare these regimes, I define an *targeting ratio* $\mathcal{R}^s$ that measures the fraction of the first-best gain (over uniform transfers) captured by regime s.

The main finding is that income targeting captures only a small fraction of the potential gains from perfect MPC targeting. In the baseline calibration with an average \$1,000 transfer budget, the information ratio for income targeting is $\mathcal{R}^y \approx 0.23$, meaning that moving from uniform to income-targeted transfers closes less than one quarter of the gap to the first best. By contrast, the information ratio for asset-and-income targeting is $\mathcal{R}^{a,y} \approx 0.89$. A single-$\beta$ benchmark strengthens this interpretation: without preference heterogeneity, (a, y)-targeting replicates the first best exactly, and even income targeting captures roughly 74% of the gains. The large drop in $\mathcal{R}^y$ when preference heterogeneity is introduced echoes the empirical finding in Lewis et al. (2026) that observables explain little MPC variation: in both cases, the variation that matters most for policy is driven by characteristics the government cannot see.

The limited effectiveness of income targeting reflects primarily the residual heterogeneity in MPCs conditional on income, driven by unobserved discount factors and the endogenous distribution of assets across types, neither of which the income-targeted planner can observe. In contrast, income is much more informative about the MPC at the household level in the single-β benchmark, in which $\mathcal{R}^y$ stays high across all different transfer budgets.

A rule-of-thumb comparison reinforces this interpretation: a policy that gives a uniform payment to every household below an income cutoff performs nearly as well as the optimal income-targeted schedule. The low $\mathcal{R}^y$ therefore reflects what income can actually reveal about the MPC.

In an extension, I introduce joint heterogeneity in discount factors and income risk. Recent evidence suggests that MPCs are correlated with earnings uncertainty (Savoia, 2023; Koşar and Melcangi, 2025), making income a better signal of the MPC.[3] Even under this favorable correlation structure, however, the targeting ratio for income targeting $\mathcal{R}^y$ is between 0.31 and 0.44, confirming that a large share of the gains from targeting remains inaccessible without observing preferences.

The paper contributes to the literature on fiscal transfers in heterogeneous-agent models (see, for example, Oh and Reis, 2012; Kaplan and Violante, 2014; Nygaard et al., 2022; Aguiar et al., 2024; Becko et al., 2025; Andreolli and Surico, 2026). A common theme in this work is that targeting high-MPC households raises aggregate consumption, but

[3]Savoia (2023) documents a negative correlation between income risk and MPC, while Koşar and Melcangi (2025) find that MPCs are increasing and concave in subjective earnings growth uncertainty.

the magnitude of the gains depends on how finely the planner can condition.[4] Oh and Reis (2012) show that targeting raises multipliers through both wealth and aggregate demand channels. Kaplan and Violante (2014) find that rebates phased out near median income deliver the strongest consumption response in a two-asset model. Nygaard et al. (2022) develop an algorithm for discrete transfers across demographic groups and apply it to the 2008 tax rebates under a consumption-maximizing planner and to the 2021 checks under an inequality-averse welfare planner; their 2008 exercise shares my objective and uses discount-factor heterogeneity to generate MPC dispersion. Becko et al. (2025) extend the analysis to a production-network economy and show that transfers should be directed toward high-MPC households, while optimal purchases policy depends on the input-output structure of production. Andreolli and Surico (2026) document that smaller transfers to a broader group outperform larger transfers to fewer recipients. Most closely related, Aguiar et al. (2024) show in a two-asset model with heterogeneity in discount factors and intertemporal elasticities of substitution that targeting the bottom net-worth quartile raises the aggregate MPC from 0.32 to 0.74, while the same exercise in a single-type model produces a much smaller gain. These papers establish the value of MPC-based targeting under specific policy rules, but leave open how much of that value is accessible to a planner constrained by the information set available in practice.

I share the emphasis of this literature on preference heterogeneity and the informational content of observables, but ask a different question: given the MPC heterogeneity present in the population, how much of the first-best targeting gain survives when the planner does not observe preferences? Rather than comparing a fixed set of targeting rules, I solve for the optimal transfer schedule within each information set and measure how much of the first-best consumption gain it recovers. The targeting ratio $\mathcal{R}^k$ thereby quantifies the targeting value of each observable set on a common scale, and is directly comparable across information sets. Relative to Nygaard et al. (2022), who provide a general algorithm for optimal discrete allocation of stimulus transfers and apply it to the 2008 and 2021 U.S. programs, my focus is the informational structure of the targeting problem itself.

[4] Boutros (2023) studies a distinct tradeoff between targeting precision and implementation delay in a two-sector HANK model; the friction I analyze is informational rather than temporal.

2 Model

2.1 Households

Time is discrete and the model is set at quarterly frequency. There is a continuum of households on $[0, 1]$. Each household is characterized by a state (a, y, β): liquid assets a, idiosyncratic productivity y, and a fixed discount factor β.

Preferences are

$$\mathrm{E}_0 \sum_{t=0}^{\infty} \beta^t \, u(c_t), \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}. \tag{1}$$

Risk aversion γ is common across households; all ex-ante heterogeneity is channeled through β.

Households receive an exogenous income process that is AR(1) in logs:

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2). \tag{2}$$

The budget constraint is then

$$c + a' = (1 + r)a + y, \quad a' \geq \underline{a}, \quad c \geq 0. \tag{3}$$

For simplicity, the discount factor β is drawn once from a two-point distribution, with types β_L and β_H, and weights ω_L and ω_H.

Each type β solves a standard consumption-savings problem:

$$V(a, y; \beta) = \max_{c, a'} \left\{ u(c) + \beta \; \mathrm{E} \left[V(a', y'; \beta) \mid y \right] \right\} \tag{4}$$

subject to the budget constraint. This yields policy functions $c(a, y; \beta)$ and $a'(a, y; \beta)$.

Three properties of the consumption function $c(a, y; \beta)$ are central to the analysis. First, c is concave in a, so the MPC $\partial c / \partial a$ is decreasing in a (Carroll and Kimball, 1996). Second, at given (a, y), a lower β implies a higher MPC. Third, combining these two properties, the average MPC out of a lump-sum transfer t is decreasing in t: as the transfer grows, the household moves along its concave consumption function, and the marginal response falls.

2.2 The Government's Problem

The government has a budget of $M > 0$ and wants to maximize the aggregate consumption response:[5]

$$\Delta C = \int [c(a+t, y; \beta) - c(a, y; \beta)]\, dF(a, y, \beta), \tag{5}$$

where t is the transfer received by a household, which depends on what the government observes, and $F(a, y, \beta)$ is the stationary joint distribution of assets, income, and discount factors. The budget constraint is:

$$\int t\, dF(a, y, \beta) = M. \tag{6}$$

I impose $t \geq 0$, meaning that the government can only give transfers but cannot tax through this program, to capture the design of actual fiscal stimulus payments such as the 2008 Economic Stimulus Payments and the 2020 Economic Impact Payments.[6] Under the optimized regimes, the government may set $t = 0$ for some households and allocate larger transfers to others, subject only to the aggregate budget constraint (6). As a result, treated households may receive transfers substantially larger than the per-capita budget M. [7]

Because the household consumption function is concave in cash on hand, the government's objective function is strictly concave in the transfer amount. The optimal allocation is therefore fully characterized by a set of first-order conditions that equate the expected marginal MPC, conditional on the planner's information set, to a common Lagrange multiplier λ. Households whose conditional expected marginal MPC at $t = 0$ falls below λ receive nothing. For the numerical solution, under each regime, an outer bisection on the Lagrange multiplier λ enforces the budget constraint, while an inner bisection at each point in the government's information partition finds the transfer that equates the (conditional expected) marginal MPC to λ.[8]

[5] Since the population has measure one, M is both the per-capita and the total transfer budget.

[6] For a mechanism design approach to fiscal stimulus with unobservable household heterogeneity, see Zhang (2023).

[7] The analysis is partial equilibrium: prices are fixed and the transfer budget M is taken as given. One interpretation is that the government finances M through debt, so the exercise isolates the first-round consumption response before any general equilibrium feedback. See Auclert et al. (2024) and Ferriere and Navarro (2024) for general equilibrium analyses where the financing of fiscal policy across households plays a central role for the size of multipliers.

[8] Details are in Appendix A.

2.2.1 Regime 1: First Best — Government Observes (a, y, β)

The government chooses a transfer function $t(a, y, \beta) \geq 0$. With full information, the first-order condition requires:

$$\frac{\partial c(a + t^{FB}, y; \beta)}{\partial a} = \lambda \quad \text{for all } (a, y, \beta) \text{ with } t^{FB}(a, y, \beta) > 0. \tag{7}$$

Since the marginal MPC is strictly decreasing in cash on hand, the optimal transfer can be obtained by inversion:

$$t^{FB}(a, y, \beta) = \max\left\{ \left(\frac{\partial c}{\partial a}\right)^{-1} (\lambda;\, y, \beta) - a,\;\; 0 \right\}, \tag{8}$$

where $(\partial c/\partial a)^{-1}(\lambda; y, \beta)$ is the asset level at which the marginal MPC equals λ for a household with productivity y and discount factor β.

2.2.2 Regime 2: Government Observes (a, y) but not β

The government chooses $t(a, y) \geq 0$. Since the transfer cannot depend on β, every household at a given (a, y) receives the same transfer regardless of type. The government's problem is:

$$\max_{t(a,y)\geq 0} \int \left[c(a + t(a, y), y; \beta) - c(a, y; \beta)\right] dF(a, y, \beta) \tag{9}$$

subject to $\int t(a, y)\, dF(a, y) = M$, where $F(a, y)$ is the marginal distribution over liquid assets and income. The first-order condition equates the conditional expected marginal MPC across all points that receive a positive transfer:

$$\int \frac{\partial c(a + t^{AY}, y; \beta)}{\partial a}\, dF(\beta \mid a, y) = \lambda \quad \text{for all } (a, y) \text{ with } t^{AY}(a, y) > 0. \tag{10}$$

The left-hand side is a weighted average of marginal MPCs across β-types at the point (a, y), with weights given by the posterior $F(\beta \mid a, y)$.

2.2.3 Regime 3: Government Observes only y

The government chooses $t(y) \geq 0$. Every household with income y receives the same transfer, regardless of assets or type. The government's problem is:

$$\max_{t(y)\geq 0} \int \left[c(a + t(y), y; \beta) - c(a, y; \beta)\right] dF(a, y, \beta) \tag{11}$$

subject to $\int t(y)\,dF(y) = M$, where $F(y)$ is the marginal distribution of income. The first-order condition is:

$$\int \frac{\partial c(a + t^Y, y; \beta)}{\partial a}\, dF(a, \beta \mid y) = \lambda \quad \text{for all } y \text{ with } t^Y(y) > 0. \tag{12}$$

The left-hand side is the average marginal MPC across all households with income y, averaging over the joint distribution of (a, β) conditional on y.

2.2.4 Regime 4: Uniform Transfers

Every household receives $t^U = M$. The aggregate consumption response is:

$$\Delta C^U = \int [c(a + M, y; \beta) - c(a, y; \beta)]\, dF(a, y, \beta). \tag{13}$$

No optimization is needed.

2.3 Measuring the Value of Information

The four regimes defined above form a natural hierarchy of information sets. To quantify how much each additional piece of information contributes to targeting, I define the *targeting ratio* for any feasible targeting regime k:

$$\mathcal{R}^k = \frac{\Delta C^k - \Delta C^U}{\Delta C^{FB} - \Delta C^U} \in [0, 1]. \tag{14}$$

The numerator is the gain in aggregate consumption from regime k relative to uniform transfers; the denominator is the maximum feasible gain, achieved under the first best. The ratio $\mathcal{R}^k$ therefore measures the fraction of the total targeting gain that regime k captures.[9]

Uniform transfers serve as the natural zero-information baseline: they require no knowledge of individual household characteristics. The ratio $\mathcal{R}^k$ therefore isolates the value of the information used by regime k by netting out the aggregate consumption response that any transfer program would generate regardless of how it is allocated, which depends on the overall level of MPCs in the population but not on the government's ability to distinguish between high and low MPC households.

Applied to the regimes above, $\mathcal{R}^y$ captures the value of observing income alone, while $\mathcal{R}^{a,y}$ captures the value of observing both assets and income. The gap $\mathcal{R}^{a,y} - \mathcal{R}^y$ isolates

[9] A related summary statistic is the resource-equivalent variation of Nygaard et al. (2022), which measures the fractional budget saving from the optimal allocation relative to an alternative.

the additional contribution of asset information beyond income. If $\mathcal{R}^{a,y}$ is itself bounded well below one, then unobservable preference heterogeneity limits the effectiveness of any targeting scheme based on observable state variables, regardless of how rich the observable information set is.

In the model, households with low discount factors accumulate little liquid wealth precisely because of their impatience, creating a strong correlation between asset holdings and β. Observing (a, y) therefore allows the planner to partially identify the household's type and approach the first-best allocation. The targeting ratio $\mathcal{R}^{a,y}$ quantifies this targeting gain and can be interpreted as an upper bound on what a government with access to household balance-sheet data could achieve relative to the first best. In practice, however, governments rarely use liquid asset holdings to determine transfer eligibility: asset data is difficult to verify in real time, and means-testing on liquid wealth raises political and administrative obstacles (Jappelli and Pistaferri, 2020; Vannutelli, 2025).

2.4 What Does the Government Need to Know?

Implementing a targeting regime requires the government to know not only the observables it conditions on, but also how the average household at each observable state responds to transfers. As the information set shrinks from (a, y, β) to (a, y) to y alone, the government loses the ability to separate individual households, yet must still know the expected consumption response $\mathrm{E}[c(a + t, y; \beta) \mid y]$ at every income level and for every transfer amount t. Uniform transfers, by contrast, require no knowledge of individual characteristics or of the underlying consumption function.

This knowledge requirement has a qualitative as well as a quantitative dimension: the direction of the targeting gains depends fundamentally on the sign of the correlation between income and the MPC. In the model, this correlation is strongly negative. Low-income households hold fewer assets and have higher MPCs, so the income-targeted government optimally tilts transfers toward them.[10]

Throughout the analysis, I assume that the government knows the model and can compute these conditional objects from the stationary distribution, so the informational friction stems purely from the inability to observe individual household states. In practice, implementing such a policy would require estimating the conditional response function from data, for example, through direct survey questions on hypothetical transfers (Jappelli

[10]This is consistent, for example, with the empirical findings of Fagereng et al. (2021) and Boehm et al. (2025). However, Lewis et al. (2026) document a positive correlation between income and the MPC in U.S. data. If this positive correlation holds in practice, income targeting would tilt transfers in the opposite direction, and the optimal transfer schedule would differ substantially from the one implied by the model.

and Pistaferri, 2014; Fuster et al., 2020; Andre et al., 2025), randomized experiments (Boehm et al., 2025), or quasi-experimental variation in transfer receipts (Fagereng et al., 2021). The reliability of such estimates depends on sample sizes, the macroeconomic stability of the underlying relationship over time, and, for survey-based measures, truthful reporting. For instance, an estimate with the wrong sign across the income distribution would reverse the direction of targeting altogether. Therefore, correctly identifying the sign of the income-MPC relationship is just as critical as recovering its exact magnitude.[11]

3 Calibration

To evaluate the quantitative importance of the government's information constraints in designing fiscal transfers, I calibrate the model to match empirical estimates of MPC dispersion and the aggregate liquid wealth distribution.

The model is set at quarterly frequency in partial equilibrium with exogenous prices. The annual risk-free rate is set to $r = 1\%$ and the coefficient of relative risk aversion to $\gamma = 2$. Quarterly labor income is normalized to have mean one.

Regarding the income process (2), the persistence parameter $\rho = 0.91^{1/4} \approx 0.977$ (quarterly) follows Floden and Lindé (2001), while the innovation standard deviation $\sigma_\varepsilon = 0.753$ follows Auclert et al. (2024). The income process is discretized using the Rouwenhorst method with 7 grid points (Kopecky and Suen, 2010).

I model ex-ante heterogeneity through two discount factor types, β_L and β_H, with population shares ω_L and $\omega_H = 1 - \omega_L$. The impatient type is fixed at $\beta_L = 0.91$, a value sufficiently low that these households hold near-zero assets in steady state and have MPCs close to one.[12] This leaves two free parameters: β_H and ω_L.

I calibrate these to match two moments. The first target is the annualized liquid wealth-to-income ratio $A/Y = 1.4$, following McKay et al. (2016). The second target is a cross-sectional standard deviation of MPCs equal to 0.357, following the survey evidence of Jappelli and Pistaferri (2014). Since the paper studies how much of this dispersion a government can exploit through targeting, disciplining the model to match the observed degree of MPC heterogeneity is essential.[13]

[11] An active literature in statistical decision theory studies how to use empirical data to design optimal policies under constraints. See, e.g., Manski (2004), Kitagawa and Tetenov (2018), and Athey and Wager (2021).

[12] Given the no-borrowing constraint, consumption behavior becomes relatively insensitive to further reductions in β_L.

[13] In the SHIW survey used by Jappelli and Pistaferri (2014), each respondent answers the MPC question for a transfer equal to one month of household income. For simplicity, I calibrate the model to match the standard deviation of the derivative of the consumption function with respect to a small increase in cash on

The calibration yields $\beta_H = 0.973$ and $\omega_L = 0.182$. The impatient type comprises roughly 18% of the population, holds essentially zero assets, and has MPCs near one, while patient households hold the vast majority of aggregate wealth. The population-weighted mean MPC is 0.24. As an untargeted check, the fraction of hand-to-mouth households (those at $a = 0$) is approximately 23%, close to the 23.3% reported by Aguiar et al. (2024) using PSID data.

Table 1 collects all parameter values. For the single-β benchmark presented in Section 4.1, I calibrate a single discount factor to match the same aggregate wealth target $A/Y = 1.4$ (annualized), holding all other parameters fixed. The standard deviation of MPCs is no longer a target in this exercise but an outcome of the model.

Finally, the per-capita transfer budget is expressed in dollars using an annual income of \$67,000, following Kaplan and Violante (2022). All four regimes are subject to the same aggregate budget constraint $\int_0^1 t_i \, di = M$, so that differences in consumption responses reflect only the allocation of transfers, not total spending. I report results for $M \in \{\$200, \$500, \$1{,}000, \$2{,}000\}$, covering a range from small targeted payments to transfers comparable to examples of stimulus transfers in the U.S., such as the 2008 Economic Stimulus Payments.

4 Results

4.1 Benchmark: No Preference Heterogeneity

Before turning to the main results, it is instructive to examine a benchmark economy with a single discount factor. In this case, all MPC variation arises from differences in assets and income, both of which are in principle observable. The cross-sectional standard deviation of MPCs is 0.281, considerably lower than in the baseline calibration with preference heterogeneity (0.357), where it is a calibration target: households are more similar to each other, so even coarse targeting schemes lose less relative to the first best.

Table 2 reports the results. Two features stand out. First, the (a, y)-targeted regime replicates the first best exactly: $\mathcal{R}^{a,y} = 1$. This is not surprising, since with a common discount factor, the pair (a, y) is a sufficient statistic for the household's MPC, so a government observing assets and income faces no informational loss. As shown below, $\mathcal{R}^{a,y}$ falls to approximately 0.89 once discount factor heterogeneity is introduced (Table 3), and this

hand. An alternative calibration target could be the standard deviation from Lewis et al. (2026), but their estimate for nondurables is particularly low (0.09), while for durables it is considerably larger (0.49) but includes households with MPCs above one, making it not directly comparable to this model. In the SHIW, the reported MPCs are by construction between zero and one.

Table 1: Baseline Calibration

Parameter	Description	Value	Source/Target
Preferences and technology			
γ	Relative risk aversion	2.0	Standard
r	Risk-free rate (quarterly)	0.0025	1% annual
$\underline{a}$	Borrowing limit	0	No borrowing
Income process			
ρ	Persistence (quarterly)	0.977	Floden and Lindé (2001)
σ_ε	Innovation std. dev.	0.753	Auclert et al. (2024)
Discount factor distribution (calibrated)			
β_L	Impatient type	0.91	Fixed
β_H	Patient type	0.973	$A/Y = 1.4$ (annual)
ω_L	Share impatient	0.182	$\text{Std}(\text{MPC}) = 0.357$

Notes: The wealth-to-income target follows the liquid wealth calibration of Kaplan and Violante (2022). The MPC standard deviation target is from Jappelli and Pistaferri (2014). The fraction of hand-to-mouth households (defined as those holding zero assets, $a = 0$) is approximately 23%.

gap is entirely attributable to unobservable preferences.[14]

Second, income targeting is substantially more effective than it will be under β heterogeneity: $\mathcal{R}^y \approx 0.74$–$0.78$. Without preference heterogeneity, income is an imperfect proxy for the MPC only because it does not reveal the household's asset position. Income remains a useful signal because, through the policy function, low-income households tend to hold fewer assets and thus have higher MPCs. Moreover, with lower MPC dispersion across the population, the cost of imperfect targeting is smaller. Adding β heterogeneity breaks this mapping: two households with the same income can have very different asset positions and MPCs depending on their patience, and the increased MPC dispersion raises the stakes of misallocation.

Finally, note that $\mathcal{R}^y$ is mildly *increasing* in M in the single-β model, whereas under β heterogeneity it drifts down over most of the range (Table 3). Income is a stronger proxy for the MPC without preference heterogeneity, and the targeting ratio remains high across all transfer sizes considered.

[14]Figures 4 and 5 in Appendix B.1 show the budget allocation and transfer schedule for this benchmark at $M = \$1{,}000$.

Table 2: Consumption Responses and Targeting Ratios: Single β Benchmark

	$M = \$200$	$M = \$500$	$M = \$1{,}000$	$M = \$2{,}000$
Aggregate MPC ($\Delta C/M$)				
First Best	0.283	0.201	0.157	0.124
(a, y)-Targeted	0.283	0.201	0.157	0.124
Income-Targeted	0.243	0.177	0.141	0.114
Uniform	0.131	0.107	0.092	0.081
Targeting Ratios				
$\mathcal{R}^{a,y}$	1.000	1.000	1.000	1.000
$\mathcal{R}^{y}$	0.735	0.741	0.755	0.777

Notes: Single discount factor calibrated to match $A/Y = 1.4$ (annualized). All other parameters as in Table 1. $\mathcal{R}^{a,y} = 1$ because (a, y) is a sufficient statistic for the MPC without heterogeneity in the discount factor β.

4.2 Consumption Responses and Targeting Ratios

Table 3 presents the central results. Under the first-best regime, where the government observes (a, y, β), the aggregate MPC out of a \$1,000 transfer is 0.28.[15] Under uniform transfers, the aggregate MPC is 0.146. The gap between these two represents the total gains from targeting.

Income targeting captures only a modest fraction of these gains. The targeting ratio $\mathcal{R}^{y}$ is 0.23 at $M = \$1{,}000$, meaning that a government observing only income can realize roughly one quarter of the first-best gains. By contrast, a government observing both assets and income achieves $\mathcal{R}^{a,y} = 0.89$, recovering nearly all the gains. Compared with the single-β benchmark in Table 2, the entire gap between $\mathcal{R}^{a,y}$ and 1.0 is due to unobservable β, and the drop in $\mathcal{R}^{y}$ from roughly 0.74 to 0.23 quantifies the additional information loss from preference heterogeneity when the government observes only income.

Table 3 also reveals the concavity channel. As M grows from \$200 to \$2,000, the first-best MPC falls from 0.50 to 0.22, and the uniform MPC falls from 0.22 to 0.12. The targeting ratio $\mathcal{R}^{y}$ drifts down over most of the range, from 0.25 at M = \$500 to 0.21 at M = \$2,000, indicating that larger transfers further erode the gains from income targeting. The intuition is the following: the consumption function is concave, so MPCs fall with transfer size for all households. As the government gives more to each recipient, the returns to concentrating transfers on high-MPC households diminish. The (a, y)-targeted ratio $\mathcal{R}^{a,y}$ is more stable, remaining between 0.87 and 0.90 across transfer sizes, because asset information continues

[15] Introducing a low-discount-factor type with MPCs near one increases the aggregate MPCs relative to the single-β benchmark of Section 4.1.

Table 3: Aggregate Consumption Responses and Targeting Ratios

	$M = \$200$	$M = \$500$	$M = \$1{,}000$	$M = \$2{,}000$
Aggregate MPC ($\Delta C/M$)				
First Best	0.503	0.358	0.280	0.220
(a, y)-Targeted	0.465	0.337	0.265	0.210
Income-Targeted	0.290	0.221	0.177	0.142
Uniform	0.220	0.174	0.146	0.122
Targeting Ratios				
$\mathcal{R}^{a,y}$	0.867	0.888	0.895	0.897
$\mathcal{R}^{y}$	0.247	0.254	0.231	0.206

Notes: $\Delta C/M$ is the aggregate consumption response per dollar transferred. $\mathcal{R}^{a,y}$ and $\mathcal{R}^{y}$ are targeting ratios as defined in equation (14).

to sort households effectively by their position on the consumption function even as the function flattens.

In absolute terms, the differences in $\Delta C/M$ across regimes also shrink as M increases. At $M = \$200$, the gap between the first best and uniform transfers is 28 cents per dollar; at $M = \$2{,}000$, it falls to 10 cents.

Figure 1 explores how the targeting ratios vary with the degree of preference heterogeneity. I recalibrate the model for a range of Std(MPC) targets between 0.30 and 0.44, holding $A/Y = 1.4$ fixed and adjusting (β_H, ω_L) at each point. Higher dispersion requires either a wider gap between β_L and β_H or a larger share of impatient households, both of which increase the mass of households whose MPCs are near one or near zero. Income cannot distinguish between these groups, so $\mathcal{R}^y$ falls monotonically as Std(MPC) increases, from roughly 0.44 at Std $= 0.30$ to about 0.20 at Std $= 0.44$. Asset-and-income targeting, by contrast, becomes more effective: $\mathcal{R}^{a,y}$ rises from 0.89 to above 0.95 over the same range. The reason is that patient households with high β_H accumulate large asset buffers and have very low MPCs, so observing assets allows the planner to identify and exclude them from the transfer pool. The wider the spread in β, the more informative assets become as a signal of type, and the larger the gap between what income targeting and asset-and-income targeting can achieve.

Figure 2 illustrates the allocation of the budget across households at $M = \$1{,}000$. The left panel shows the distribution of MPCs in the stationary distribution; the right panel shows the share of the total budget allocated to each group under each regime. Under the first best, about 80% of the budget goes to households with MPCs close to one, while uniform transfers mechanically allocate spending in proportion to population shares.

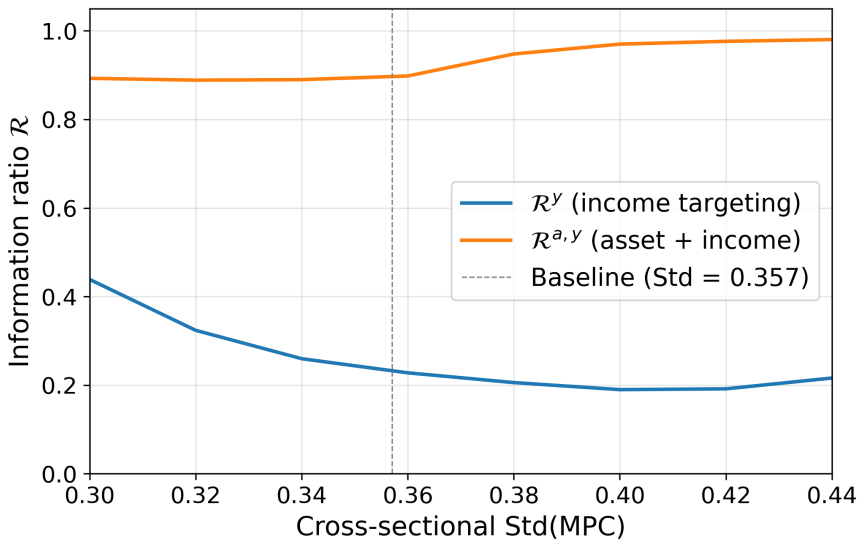

Figure 1: Targeting Ratios and the Degree of Preference Heterogeneity ($M = \$1{,}000$)

Notes: Each point recalibrates (β_H, ω_L) to match the indicated Std(MPC) target while holding $A/Y = 1.4$ (annualized) and $\beta_L = 0.91$ fixed. The transfer budget is $M = \$1{,}000$. The dashed line marks the baseline calibration (Std $= 0.357$).

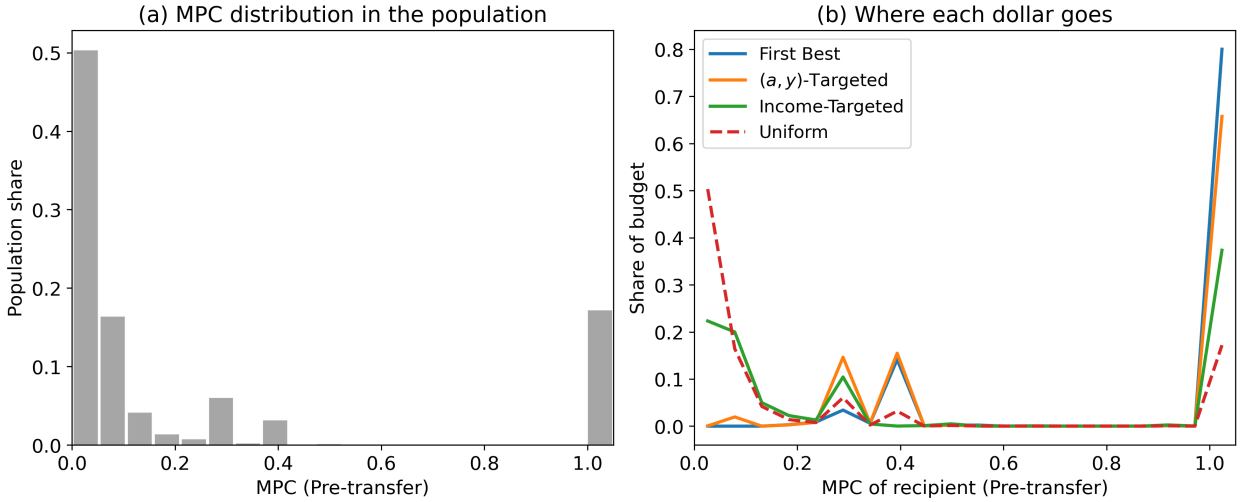

Figure 2: MPC Distribution and Budget Allocation across Regimes ($M = \$1{,}000$)

Notes: The left panel shows the cross-sectional distribution of MPCs in the stationary distribution. The right panel shows the share of the total budget allocated to each MPC group under each targeting regime.

The (a, y)-targeted regime closely tracks the first-best allocation, confirming that asset information is highly effective at identifying high-MPC households. The income-targeted regime shifts some mass toward low-income households but cannot distinguish between high-MPC and low-MPC households at the same income level.

Figure 3 shows the optimal income-targeted transfer schedule $t(y)$ alongside the population density across income levels. The schedule is steeply decreasing in income, reflecting

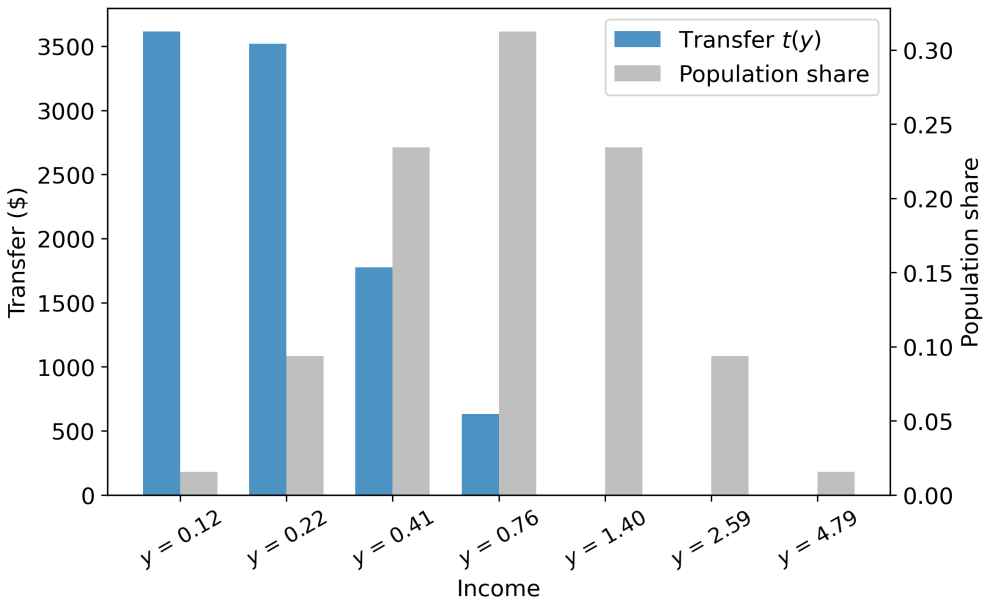

Figure 3: Optimal Income-Targeted Transfer Schedule and Population Density (M = \$1,000)

Notes: The blue bars (left axis) show the optimal transfer $t(y)$ as a function of income under income targeting. The gray bars (right axis) show the population density across income levels in the stationary distribution.

the negative correlation between income and the MPC in the model. More broadly, all targeting regimes that improve on uniform transfers imply significant concentration of the budget. At M = \$1,000, the first-best allocation directs transfers to roughly 28% of the population, with treated households receiving on average approximately \$3,500 each. Even income targeting, which captures only a quarter of the first-best gains, concentrates the budget on about 66% of households, with per-recipient transfers averaging \$1,500. In both cases, the planner concentrates resources on households with the highest MPCs and excludes the rest.

4.3 Rule-of-Thumb Targeting

The income-targeted regime assumes the government can solve for the optimal transfer schedule $t(y)$ at every income level. Therefore, it is informative to study transfer programs that use simpler rules: every household below a given income cutoff receives the same payment, and those above receive nothing. This subsection evaluates such rule-of-thumb policies and asks whether the simplicity of the instrument entails a significant loss relative to the fully optimized schedule.

The government chooses an income threshold y^* and gives every household with $y \leq y^*$ a uniform payment $\bar{t} = M/F(y^*)$, where $F(y^*)$ is the population share below

Table 4: Rule-of-Thumb vs. Optimal Income Targeting

	$M = \$200$	$M = \$500$	$M = \$1{,}000$	$M = \$2{,}000$
Aggregate MPC ($\Delta C/M$)				
Bottom 34%	0.277	0.209	0.167	0.135
Bottom 66%	0.266	0.206	0.168	0.137
Income-Targeted (optimal)	0.290	0.221	0.177	0.142
Uniform	0.220	0.174	0.146	0.122
Targeting Ratios				
$\mathcal{R}^{34\%}$	0.201	0.190	0.157	0.133
$\mathcal{R}^{66\%}$	0.163	0.174	0.164	0.153
$\mathcal{R}^{y}$	0.247	0.254	0.231	0.206

Notes: The first two rows report $\Delta C/M$ under a policy that gives a uniform payment $\bar{t} = M/F(y^*)$ to all households below the indicated income cutoff. At $M = \$1{,}000$, the per-recipient transfer is approximately \$2,900 for the bottom-34% cutoff and \$1,500 for the bottom-66% cutoff. The last row of each panel reproduces benchmarks from Table 3. Targeting ratios are defined in equation (14).

the cutoff. Households above the threshold receive nothing. I evaluate two cutoffs corresponding to treating roughly the bottom 34% and 66% of the income distribution. The narrower cutoff concentrates the budget on households in the lower third of the income distribution, while the broader cutoff extends eligibility to the median household and beyond.

Table 4 presents the results alongside the optimal income-targeted and uniform benchmarks from Table 3. Several findings emerge.

At small budgets ($M = \$200$), the narrower cutoff (bottom 34%) outperforms the broader one: $\Delta C/M = 0.277$ versus 0.266. Both cutoffs generate consumption responses above uniform transfers (0.220) and close to optimal income targeting (0.290). The narrower cutoff performs better because, at small M, the per-recipient transfer remains modest and low-income households have higher average MPCs. Concentrating the budget on this group is therefore effective.

As the budget grows, however, the two cutoffs converge and eventually the broader cutoff dominates. At $M = \$1{,}000$, the bottom-66% cutoff yields $\Delta C/M = 0.168$, slightly above the bottom-34% cutoff at 0.167. At $M = \$2{,}000$, the gap widens: 0.137 versus 0.135. The shift reflects the concavity of the consumption function. Under the narrower cutoff, the per-recipient transfer grows proportionally faster as M increases (since the eligible population is smaller), pushing recipients further along the flat region of their consumption functions and reducing the marginal response. The broader cutoff spreads the budget

across more households, keeping per-recipient transfers smaller and preserving higher marginal responses. This pattern is consistent with the finding of Andreolli and Surico (2026) that, for a given fiscal outlay, smaller payments to more households can generate a stronger aggregate response than larger payments to fewer recipients.

Across all values of M, the best rule-of-thumb policy generates targeting ratios close to those of optimal income targeting: at $M = \$1{,}000$, $\mathcal{R}^{66\%} = 0.16$ compared with $\mathcal{R}^{y} = 0.23$ under the fully optimized schedule, and the gap in aggregate MPCs is less than one cent per dollar (0.168 versus 0.177). The gains from fine-tuning $t(y)$ are therefore quantitatively small once the government has chosen a reasonable eligibility threshold. At the same time, both remain far from the first-best aggregate MPC of 0.280, confirming that the main obstacle to effective targeting is the government's limited ability to identify high-MPC households, not the design of the transfer schedule.

4.4 Extension: Joint Heterogeneity in Preferences and Income Risk

The baseline model assumes that all households face the same income process. In practice, households differ not only in their discount factors but also in the amount of income risk they face. Recent empirical work by Koşar and Melcangi (2025) shows that the MPC is increasing and concave in subjective earnings uncertainty, while Savoia (2023) documents a negative correlation between income risk and the MPC using Italian micro-data. This motivates an extension in which households differ along two dimensions: subjective discount factor (β) and income volatility (σ_ε).

I extend the baseline model by introducing two values of the income innovation standard deviation, $\sigma^L < \sigma^H$, while maintaining a common persistence parameter ρ. Each household is now characterized by a pair $(\beta, \sigma_\varepsilon)$ drawn from a joint distribution.[16] The joint distribution over $(\beta, \sigma_\varepsilon)$ is parameterized so that the marginal distribution over σ_ε is fixed at $(0.5, 0.5)$, ensuring that average income risk is identical across all scenarios and equal to the baseline value $\sigma_\varepsilon = 0.753$. I set $\sigma^L = 0.654$ and $\sigma^H = 0.854$.[17] The correlation between β and σ_ε is controlled by a single parameter δ that shifts mass along the diagonal

[16]Since different σ_ε types face different income processes, the discretization creates grids with different supports. The income-targeted planner conditions on the household's income grid position $i_y \in \{1, \ldots, n_y\}$, choosing a common transfer $t(i_y)$ for all households at the same grid index regardless of their σ_ε type. This is equivalent to a planner who observes income rank but cannot distinguish whether a household is in a high-risk or low-risk occupation. I refer to this regime simply as income targeting throughout.

[17]The 2×2 joint probability table is constructed by starting from the product of the two marginals, $\omega_{\beta,i} \cdot \omega_{\sigma,j}$, and then adding a parameter δ to the diagonal entries and subtracting it from the off-diagonal entries. The parameter δ controls the correlation: $\delta = 0$ yields independence, $\delta < 0$ assigns impatient types disproportionately to high income risk, and $\delta > 0$ does the opposite. Non-negativity of all entries requires $|\delta| \leq \min(\omega_L, 1 - \omega_L) \cdot 0.5$.

Table 5: Calibrated Parameters under Joint $(\beta, \sigma_\varepsilon)$ Heterogeneity

Scenario	$\text{Corr}(\beta, \sigma_\varepsilon)$	β_H	ω_L	A/Y	Std(MPC)
Negative ($\delta = -0.02$)	-0.16	0.971	0.071	1.40	0.357
Positive ($\delta = +0.02$)	$+0.17$	0.970	0.056	1.40	0.357

Notes: $\beta_L = 0.91$ is fixed across scenarios. ω_L denotes the population share of the impatient type. Income volatilities are $\sigma^L = 0.654$ and $\sigma^H = 0.854$, symmetric around the baseline value $\sigma_\varepsilon = 0.753$. The marginal distribution over σ_ε is fixed at $(0.5, 0.5)$ in all scenarios, so that population-average income risk is identical across exercises.

of the joint distribution without altering either marginal.

I consider two opposite scenarios. In the *negative correlation* case ($\delta = -0.02$), impatient households are more likely to face high income risk, consistent with the findings of Koşar and Melcangi (2025). In the *positive correlation* case ($\delta = +0.02$), impatient households are instead concentrated in low-risk occupations. In the calibration, these values of δ imply $\text{Corr}(\beta, \sigma_\varepsilon) = -0.16$ and $+0.17$, respectively.[18]

To ensure comparability across scenarios, I fix $\beta_L = 0.91$ as before and recalibrate (β_H, ω_L) for each value of δ to match the same aggregate targets: a liquid wealth-to-income ratio of $A/Y = 1.4$ (annualized), and a cross-sectional standard deviation of MPCs of 0.357. Table 5 reports the calibrated parameters.

Table 6 presents the results under two variants of income targeting. In the first, labeled "Income-Targeted," the planner observes the household's income grid position and chooses a transfer schedule $t(y)$, pooling across both β and σ_ε types at each income level. This is the natural counterpart of the income-targeted regime in the baseline model: the planner sees income but has no information about preferences or income risk. In the second variant, labeled "(σ, y)-Targeted," the planner additionally observes the household's σ_ε type and can condition transfers on both income and income risk. Economically, this corresponds to a government that observes not only a household's earnings but also its occupation or sector (e.g., public employee vs. freelance contractor), which serves as a proxy for earnings volatility.

Several findings emerge. First, income targeting alone captures a larger share of the first-best gains than in the baseline model. Under negative correlation, $\mathcal{R}^y$ ranges from 0.38 to 0.44, while under positive correlation it falls to 0.31 to 0.37. Both exceed the baseline range of roughly 0.21 to 0.25. This reflects the fact that, with two distinct income processes, income rank is more informative in identifying high-MPC households.

[18]Setting $\delta = 0$ (independence) yields results intermediate between the two cases reported here. At $M = \$200$, the targeting ratios under independence are $\mathcal{R}^y = 0.40$ and $\mathcal{R}^{\sigma,y} = 0.49$.

Table 6: Consumption Responses and Targeting Ratios: Joint $(\beta, \sigma_\varepsilon)$ Model

	Negative (Corr $= -0.16$)				Positive (Corr $= +0.17$)			
	\$200	\$500	\$1,000	\$2,000	\$200	\$500	\$1,000	\$2,000
Aggregate MPC ($\Delta C/M$)								
First Best	0.403	0.285	0.221	0.175	0.442	0.310	0.239	0.186
(a, y)-Targeted	0.389	0.275	0.214	0.169	0.422	0.297	0.229	0.179
(σ, y)-Targeted	0.287	0.208	0.164	0.130	0.309	0.223	0.175	0.138
Income-Targeted	0.276	0.202	0.160	0.128	0.276	0.202	0.160	0.129
Uniform	0.174	0.140	0.118	0.100	0.180	0.145	0.123	0.103
Targeting Ratios								
$\mathcal{R}^{a,y}$	0.937	0.935	0.932	0.927	0.924	0.922	0.920	0.916
$\mathcal{R}^{\sigma,y}$	0.492	0.469	0.443	0.401	0.492	0.468	0.447	0.421
$\mathcal{R}^{y}$	0.443	0.427	0.407	0.377	0.366	0.343	0.323	0.309

Notes: Each column reports results for a per-capita transfer budget of M dollars. Annual income is \$67,000. $\Delta C/M$ is the aggregate consumption response per dollar transferred. Income-Targeted: planner observes income grid position i_y and pools across all $(\beta, \sigma_\varepsilon)$ types at each level. (σ, y)-Targeted: planner additionally observes σ_ε and optimizes separately within each risk group. $\mathcal{R}^{a,y}$, $\mathcal{R}^{\sigma,y}$, and $\mathcal{R}^{y}$ are targeting ratio as defined in equation (14). The model is recalibrated for each correlation scenario to match $A/Y = 1.4$ (annualized) and $\text{Std}(\text{MPC}) = 0.357$. $\beta_L = 0.91$ is fixed.

Second, the value of additionally observing income risk depends on the sign of the correlation between patience and income risk. The key comparison is the gap $\mathcal{R}^{\sigma,y} - \mathcal{R}^{y}$, which measures how much the planner gains from distinguishing high-risk and low-risk households at the same income rank. Under negative correlation, this gap is modest: about 5 percentage points at $M = \$200$ and 2.5 percentage points at $M = \$2{,}000$. Under positive correlation, the same gap widens to 13 percentage points at $M = \$200$ and 11 percentage points at $M = \$2{,}000$.

The intuition is as follows. The (σ, y)-targeted planner gains over the income-only planner by setting different transfer schedules for the two risk groups. These gains are larger when the two groups differ more in their average MPCs. Under positive correlation, impatient households concentrate in the low-σ group, where they face little earnings risk and therefore accumulate almost no precautionary wealth. This pulls the low-risk group's average MPC sharply above the high-risk group's, creating a large between-group gap that the (σ, y)-planner exploits. Under negative correlation, impatient households instead land in the high-σ group, where the precautionary savings motive partially offsets their impatience: they hold more assets and have lower MPCs than they would under low risk. This compresses the between-group MPC gap, leaving less scope for the (σ, y)-planner to improve upon income-only targeting.

Third, asset-and-income targeting continues to capture over 91% of the first-best gains across all scenarios. The concavity channel is also visible: as M increases, $\mathcal{R}^{y}$ and $\mathcal{R}^{\sigma,y}$ decline and all regimes converge toward a common, lower $\Delta C/M$.

5 Discussion and Policy Implications

The quantitative results show that while targeted fiscal transfers can substantially boost aggregate consumption, realizing these gains requires conditioning on household information richer than income alone. It is therefore instructive to compare these findings with related exercises in the recent literature.

Two recent papers conduct related fiscal experiments. Aguiar et al. (2024) compare aggregate MPCs under untargeted transfers, net-worth targeting, and income targeting in a multi-type two-asset model. They find that targeting the bottom quartile by net worth generates an aggregate MPC of 0.74, more than double the 0.32 from untargeted transfers, while targeting by income is far less effective (0.42). This reflects the fact that wealth is a much more informative targeting variable than income. The key difference is that Aguiar et al. (2024) assign a fixed transfer to a fixed group, whereas I solve for the optimal transfer schedule across the entire distribution. The targeting ratio $\mathcal{R}^{a,y} \approx 0.89$ in my framework provides a complementary measure of how much of the first-best gains wealth-based targeting can capture.

Andreolli and Surico (2026) use Italian survey data to show that smaller transfers distributed to a larger share of low-cash-on-hand households generate a stronger aggregate consumption response than larger transfers concentrated on fewer households. Their "less is more" result arises because borrowing constraints make the MPC out of small windfalls higher than out of large windfalls among liquidity-poor households, while non-homothetic preferences generate the opposite pattern among affluent families. A common theme across their analysis and mine is that realizing the gains from MPC heterogeneity places strong informational demands on the government. Their fiscal experiments require the planner to observe cash on hand and to select which households fall below a given percentile of its distribution, effectively conditioning on both income and financial wealth. In my framework, this corresponds to the (a, y)-targeted regime, which captures roughly 89 percent of the first-best gains.

Three policy implications follow from the quantitative analysis. First, income targeting is not very effective when there is large heterogeneity in MPCs driven by unobservable preferences and the asset positions they generate: the government captures less than a quarter of the first-best gains in the baseline calibration. Second, conditioning on a richer

set of observables, such as liquid assets, can be highly effective, recovering roughly 89% of the first-best gains. Third, the value of additional information increases when the source of heterogeneity is itself correlated with observables, as the joint $(\beta, \sigma_\varepsilon)$ extension illustrates: observing income risk raises the targeting ratio by 5 to 13 percentage points depending on the correlation structure, though even the best-case ratio of roughly 0.49 remains well below the 0.89 achieved by conditioning on assets. These findings are subject to the usual caveats of partial equilibrium analysis: they capture only the first-round consumption response and abstract for instance from prices feedback, labor supply responses, and how the government finances these transfers.

6 Conclusion

This paper quantifies the limits of targeted fiscal transfers when the government cannot observe household preferences. In an Aiyagari model with discount factor heterogeneity, income targeting captures only about one quarter of the first-best gains over uniform transfers, because income reveals little about the MPC variation driven by unobservable discount factors and the asset positions they generate. The targeting ratio $\mathcal{R}^y$ tends to fall as the transfer budget grows, so the limits of income targeting become more pronounced for larger transfers. A single-discount-factor benchmark confirms this interpretation: without preference heterogeneity, income targeting captures roughly three quarters of the first-best gains, and asset-and-income targeting replicates the first best exactly. An extension with joint heterogeneity in discount factors and income risk shows that even when income is a better signal of the MPC, income targeting captures less than half the first-best gains, confirming the robustness of the baseline finding.

Several limitations should be noted. The analysis is partial equilibrium: it abstracts from price feedback, labor supply responses, and the fiscal cost of financing transfers. It also abstracts from behavioral responses by households to the targeting rule itself: in principle, households might adjust savings or reported income in response to known targeting criteria.

The findings suggest caution about the practical gains from fiscal targeting for demand stabilization. The existing literature has emphasized MPC heterogeneity as a key determinant of fiscal multipliers, but realizing the gains from this heterogeneity requires information that governments typically lack. Conditioning on liquid wealth can recover roughly 89% of the first-best gains, but even this requires balance-sheet data that is difficult to verify and rarely used in transfer eligibility. Furthermore, all regimes that improve on uniform transfers concentrate the budget on a subset of the population: under the first

best, fewer than 30% of households receive a transfer, implying a degree of concentration that may be difficult to implement in practice regardless of the available information.

The gap between the MPC heterogeneity that exists in the population and the share a government can exploit is large, and closing it demands information well beyond what standard fiscal instruments currently use.

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A Computational Details

The household problem is solved using the endogenous grid point method of Carroll (2006) on an asset grid of 1,000 points over $[0, 15{,}000]$ (in units of quarterly income). The marginal MPC $\partial c / \partial a$ is computed using numerical differentiation on the converged policy function.

The optimal transfer allocation for Regimes 1 through 3 exploits the fact that the government's objective is concave in the transfer t. The first-order condition equates the (conditional expected) marginal MPC to a Lagrange multiplier λ across all cells that receive a positive transfer. Cells whose marginal MPC at $t = 0$ falls below λ receive nothing.

The algorithm proceeds in two nested loops. The outer loop searches over λ by bisection. For a candidate λ, the inner loop visits each cell in the government's information partition (a single (a, y, β) triple for the first best, an (a, y) pair for asset-and-income targeting, or an income level y for income targeting) and finds the transfer t^* such that the weighted average marginal MPC in that cell equals λ, again by bisection on t. If the weighted average marginal MPC at $t = 0$ is already below λ, the cell is untreated ($t^* = 0$). After solving all cells, total spending $S = \sum t^* \mu$ is computed. If $S > M$, the threshold λ is too low and the lower bound is raised. If $S < M$, the upper bound is lowered. The outer bisection terminates when $|\lambda^{hi} - \lambda^{lo}| < 10^{-10}$.

For the joint $(\beta, \sigma_\varepsilon)$ extension, the same procedure applies with four household types instead of two. The (σ, y)-targeted regime runs the inner bisection separately within each σ_ε group, while the income-targeted regime pools across all types at each income level.

B Additional Figures

B.1 Single-β Benchmark

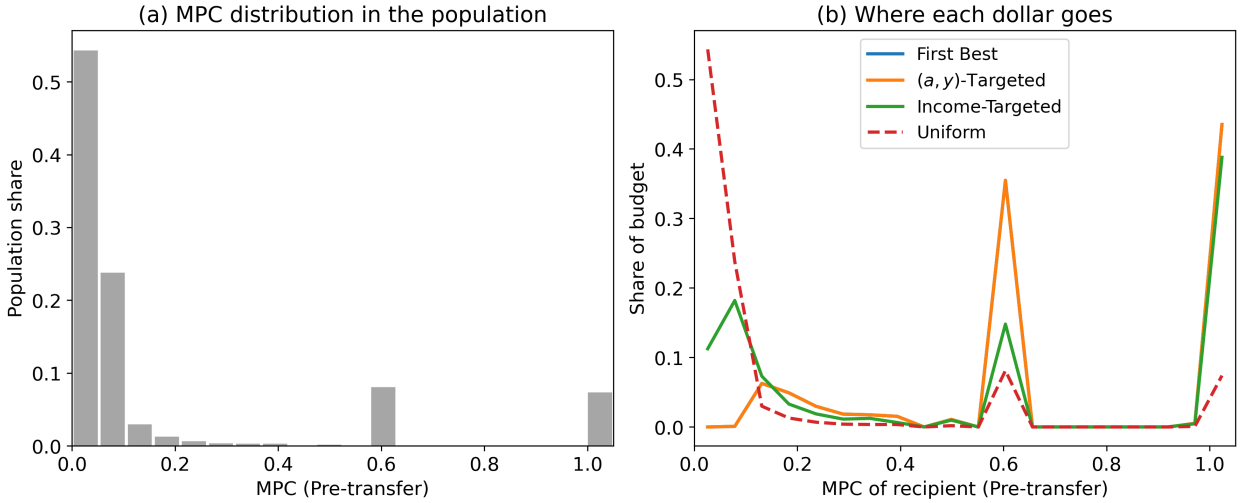

Figure 4: MPC Distribution and Budget Allocation: Single-β Benchmark ($M = \$1{,}000$)

Notes: The left panel shows the cross-sectional distribution of MPCs in the stationary distribution under a single discount factor. The right panel shows the share of the total budget allocated to each MPC group under each targeting regime. See Section 4.1 for details on the benchmark calibration.

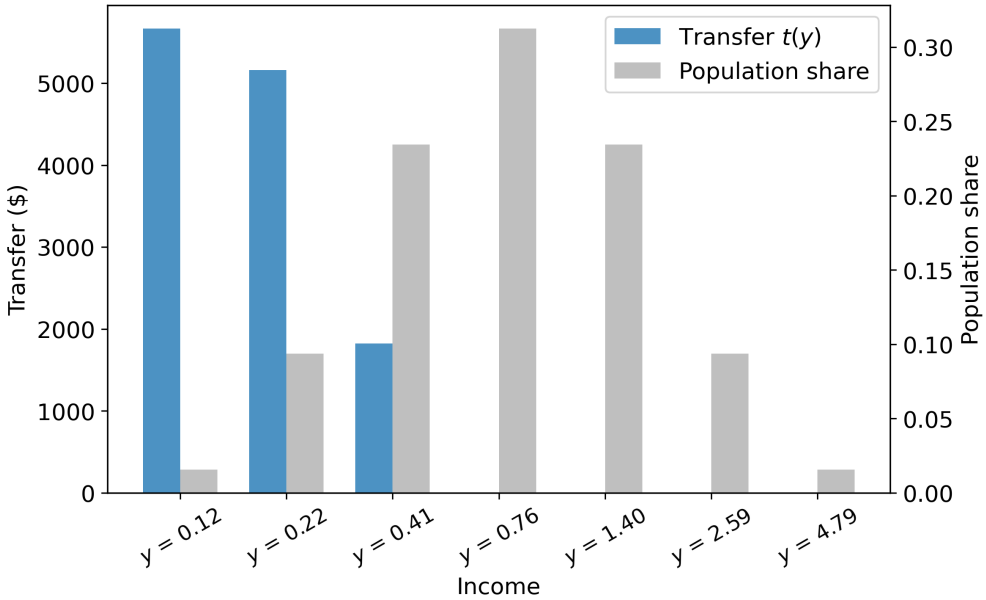

Figure 5: Optimal Income-Targeted Transfer Schedule and Population Density: Single-β Benchmark ($M = \$1{,}000$)

Notes: The blue bars (left axis) show the optimal transfer $t(y)$ as a function of income under income targeting. The gray bars (right axis) show the population density across income levels in the stationary distribution. See Section 4.1 for details on the benchmark calibration.

B.2 Joint $(\beta, \sigma_\varepsilon)$ Model

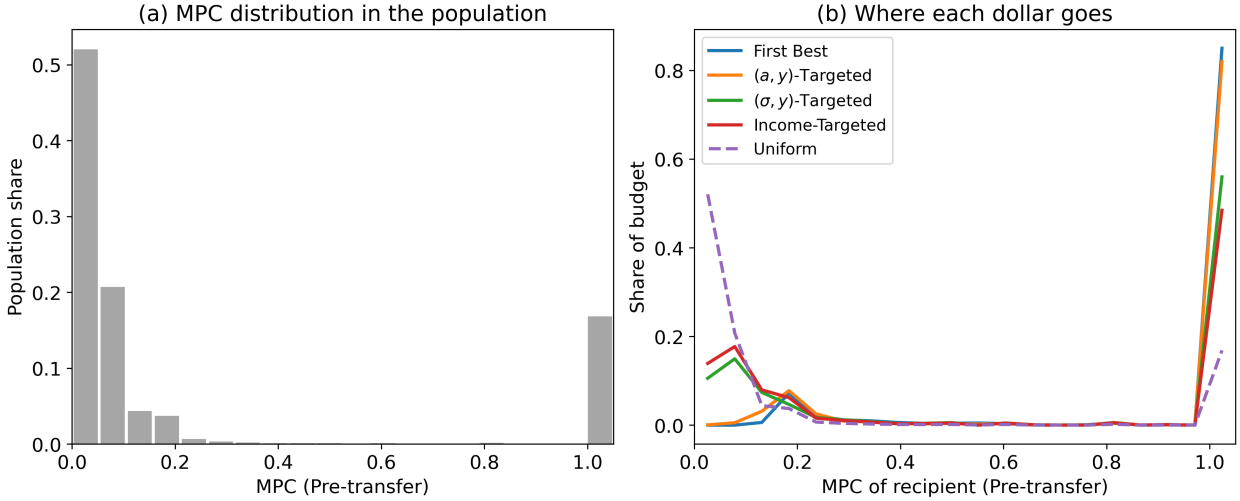

Figure 6: MPC Distribution and Budget Allocation: Negative Correlation ($M = \$1{,}000$)

Notes: The left panel shows the cross-sectional distribution of MPCs in the stationary distribution. The right panel shows the share of the total budget allocated to each MPC group under each targeting regime. $\text{Corr}(\beta, \sigma_\varepsilon) = -0.16$.

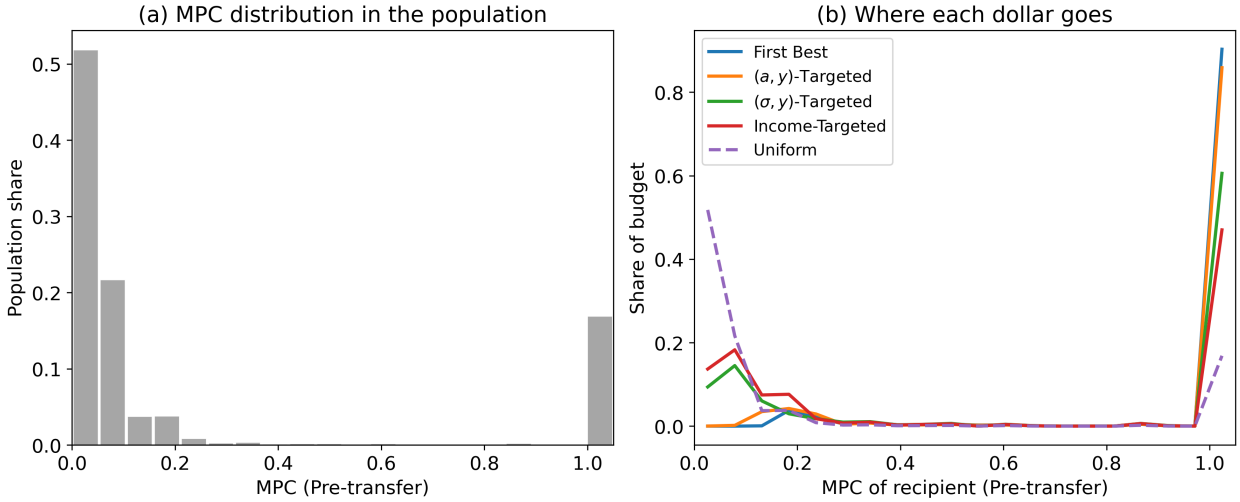

Figure 7: MPC Distribution and Budget Allocation: Positive Correlation ($M = \$1{,}000$)

Notes: The left panel shows the cross-sectional distribution of MPCs in the stationary distribution. The right panel shows the share of the total budget allocated to each MPC group under each targeting regime. $\text{Corr}(\beta, \sigma_\varepsilon) = +0.17$.

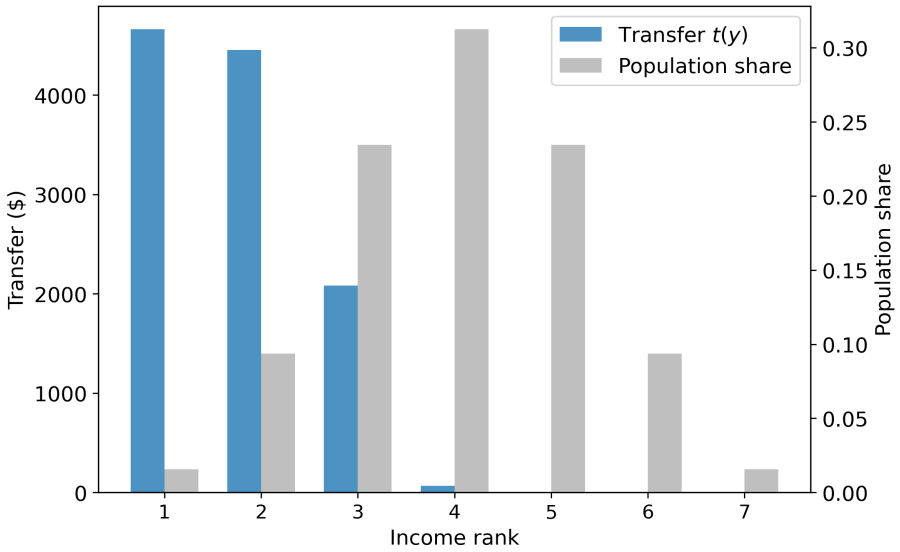

Figure 8: Optimal Income-Targeted Transfer Schedule: Negative Correlation ($M = \$1{,}000$)

Notes: The blue bars (left axis) show the optimal transfer $t(y)$ as a function of income rank under income targeting. The gray bars (right axis) show the population density across income ranks in the stationary distribution. $\text{Corr}(\beta, \sigma_\varepsilon) = -0.16$.